SIMATS
ENGINEERING

SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

CSA1401-COMPILER DESIGN

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PRACTICALS

INDEX

EXPERIMENT 1: Lexical Analyzer – identifiers, constants and operators

AIM

A lexical analyzer that ignores redundant spaces, tabs, new lines and comments, and identifies identifiers, constants and operators in a C program.

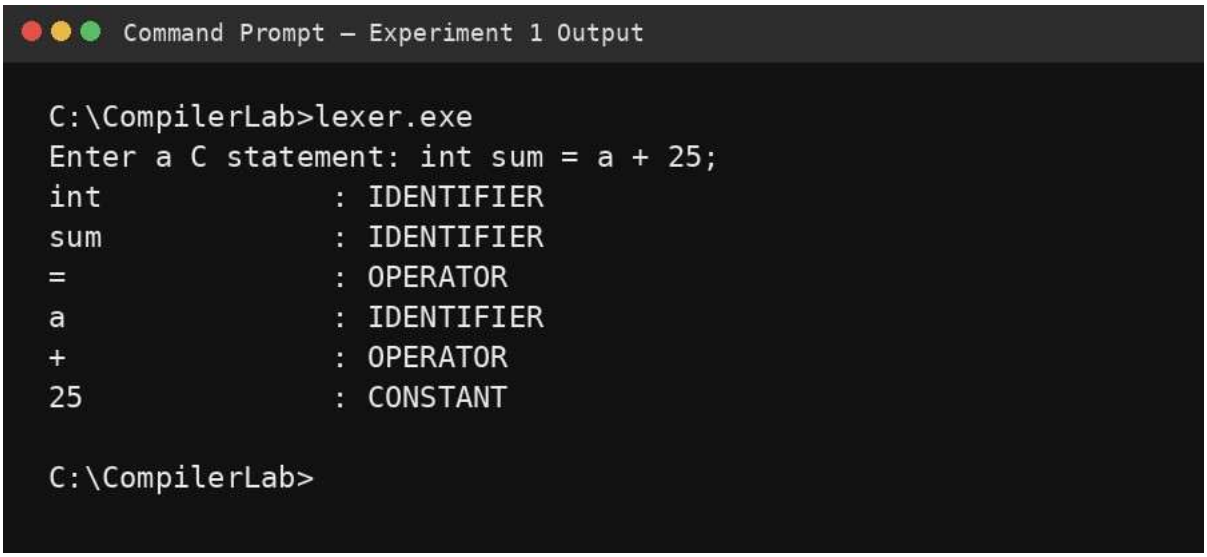
ALGORITHM

- Read the source program character by character.
- Skip spaces, tabs, new lines and comments.
- If a letter or underscore is found, collect the complete identifier.
- If a digit is found, collect the complete numeric constant.
- If an operator character is found, classify it as an operator.
- Print each recognized token with its category.

PROGRAM

```
#include <stdio.h>
#include <ctype.h>
#include <string.h>


int main(void) {
char s[1000];
   int i = 0;
   printf("Enter a C statement: ");
fgets(s, sizeof(s), stdin);


   while (s[i] != '\0') {
      if (isspace((unsigned char)s[i])) { i++; continue; }


      if (isalpha((unsigned char)s[i]) || s[i] == '_') {
char tok[100];            int j = 0;
         while (isalnum((unsigned char)s[i]) || s[i] == '_')
tok[j++] = s[i++];
         tok[j] = '\0';
         printf("%-15s : IDENTIFIER\n", tok);
} else if (isdigit((unsigned char)s[i])) {
char tok[100];            int j = 0;
         while (isdigit((unsigned char)s[i]) || s[i] == '.')
tok[j++] = s[i++];
         tok[j] = '\0';
         printf("%-15s : CONSTANT\n", tok);
} else if (strchr("+-*/%=<>!", s[i])) {
         printf("%-15c : OPERATOR\n", s[i++]);
      } else {
i++;
      }   }
return 0;
}
```

OUTPUT



RESULT

Thus, the lexical analyzer successfully identified identifiers, constants and operators while ignoring redundant white space.

EXPERIMENT 2: Lexical Analyzer – single-line and multi-line comments

AIM

To determine whether a given C source line is a single-line comment, a multi-line comment, or ordinary code.

ALGORITHM

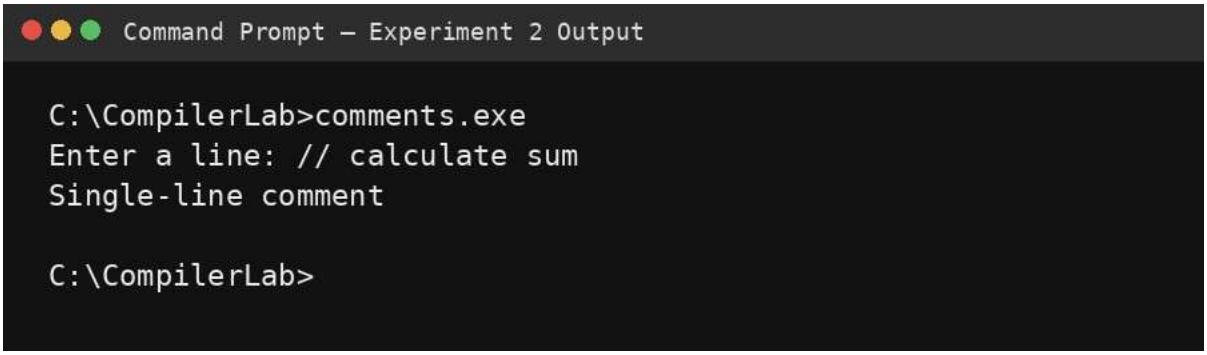
- Remove leading white space conceptually for classification.
- If the line starts with //, classify it as a single-line comment.
- If the line starts with /*, search for */ and classify it as a multi-line comment when present.
- Otherwise classify it as a non-comment line.
- Display the classification.

PROGRAM

```
#include <stdio.h>
#include <string.h>

int main(void) {     char
s[500];     printf("Enter a
line: ");
   fgets(s, sizeof(s), stdin);
```

```
   if (strncmp(s, "//", 2) == 0)
printf("Single-line comment\n");
else if (strncmp(s, "/*", 2) == 0) {
if (strstr(s + 2, "*/"))
        printf("Multi-line comment\n");
else
        printf("Start of a multi-line comment\n");
   } else
      printf("Not a comment\n");
return 0;
}
```

OUTPUT



RESULT

Thus, the program correctly identifies the specified C comment forms.

EXPERIMENT 3: Lexical Analyzer – ignore spaces, tabs, new lines and comments

AIM

To design a lexical analyzer that removes redundant white space and comments before processing the remaining source characters.

ALGORITHM

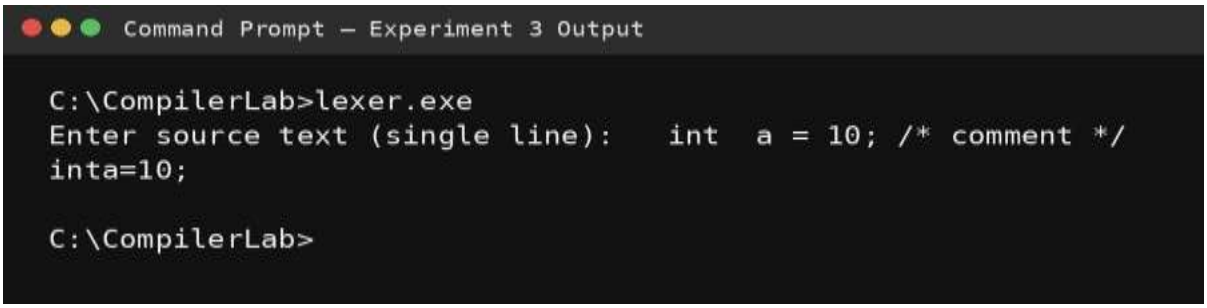
- Read the complete input text.
- Skip white-space characters.
- When // is encountered, skip until the newline.
- When /* is encountered, skip until */.
- Print all remaining characters in their original order.
- Verify that comments and redundant white space are absent.

PROGRAM

```
#include <stdio.h>
```

```
int main(void) {     char
s[1000];     int i = 0,
in_block = 0;
   printf("Enter source text (single line): ");
fgets(s, sizeof(s), stdin);


   while (s[i]) {        if (!in_block && s[i]=='/' &&
s[i+1]=='/') break;        if (!in_block && s[i]=='/' &&
s[i+1]=='*') {
          in_block = 1; i += 2; continue;
       }
       if (in_block && s[i]=='*' && s[i+1]=='/') {
          in_block = 0; i += 2; continue;
       }
       if (!in_block && s[i]!='\n' && s[i]!='\t' && s[i]!=' ')
putchar(s[i]);        i++;     }
   putchar('\n');
   return 0;
}
```

OUTPUT



RESULT

Thus, redundant spaces and comments were ignored successfully.

AIM

EXPERIMENT 4: Lexical Analyzer – validate arithmetic operators

To recognize and validate the arithmetic operators +, -, * and /.

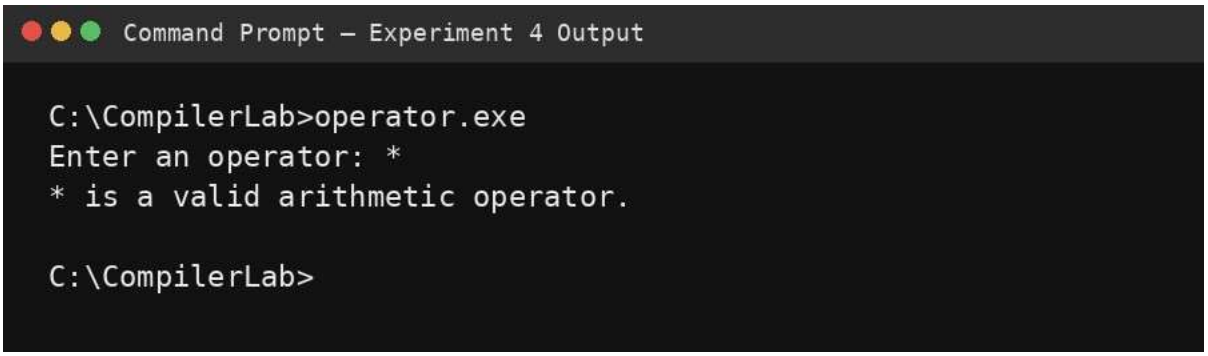
ALGORITHM

- Read an input character.
- Compare it with +, -, * and /.
- If a match is found, print the corresponding arithmetic operator.
- Otherwise report that it is not one of the required operators.

PROGRAM

```
#include <stdio.h>

int main(void) {
char ch;
   printf("Enter an operator: ");
scanf(" %c", &ch);

   if (ch == '+' || ch == '-' || ch == '*' || ch == '/')
printf("%c is a valid arithmetic operator.\n", ch);     else
       printf("%c is not a valid arithmetic operator.\n", ch);
return 0;
}
```

OUTPUT



RESULT

Thus, the specified regular arithmetic operators were validated successfully.

EXPERIMENT 5: Lexical Analyzer – count whitespaces and newlines

To count the number of whitespace characters and newline characters in an input text.

AIM

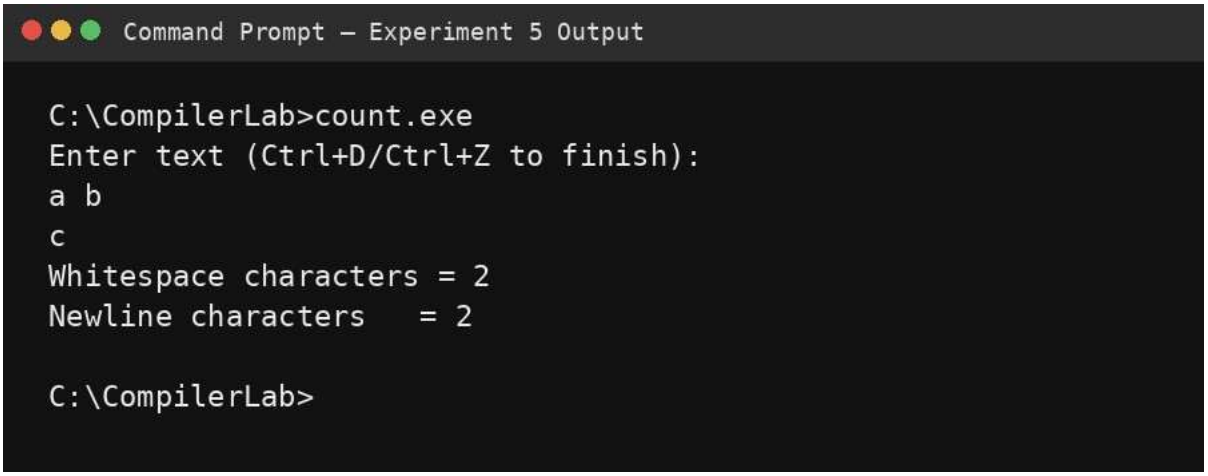
ALGORITHM

- Read characters until end of file.
- Increment the whitespace count for space, tab and other white-space characters.
- Increment the newline count whenever '\n' is read.
- Display both counts.

PROGRAM

```
#include <stdio.h>
#include <ctype.h>


int main(void) {
   int c, whitespace = 0, newline = 0;
   printf("Enter text (Ctrl+D/Ctrl+Z to finish):\n");
while ((c = getchar()) != EOF) {         if
(isspace(c)) whitespace++;
      if (c == '\n') newline++;
   }
   printf("Whitespace characters = %d\n", whitespace);
printf("Newline characters   = %d\n", newline);
return 0; }
```

OUTPUT



RESULT

Thus, the number of whitespace and newline characters was counted correctly.

EXPERIMENT 6: Lexical Analyzer – validate identifiers

To test whether a given string is a valid C-style identifier.

AIM

ALGORITHM

- Check that the first character is a letter or underscore.
- Check that all remaining characters are letters, digits or underscores.
- If both conditions hold, accept the identifier.
- Otherwise reject it.

PROGRAM

```
#include <stdio.h>
#include <ctype.h>

int main(void) {     char
s[100];     int i, valid = 1;
printf("Enter identifier: ");
scanf("%99s", s);

   if (!(isalpha((unsigned char)s[0]) || s[0] == '_'))
valid = 0;

   for (i = 1; s[i] && valid; i++)
      if (!(isalnum((unsigned char)s[i]) || s[i] == '_'))
valid = 0;

   printf("%s\n", valid ? "Valid identifier" : "Invalid identifier");
return 0;
}
```

OUTPUT

RESULT

Thus, the given identifier was validated according to the stated lexical rules.

EXPERIMENT 7: FIRST( ) – predictive parser

To compute FIRST sets for the grammar S → AaAb / BbBa, A → ε, B → ε.

AIM

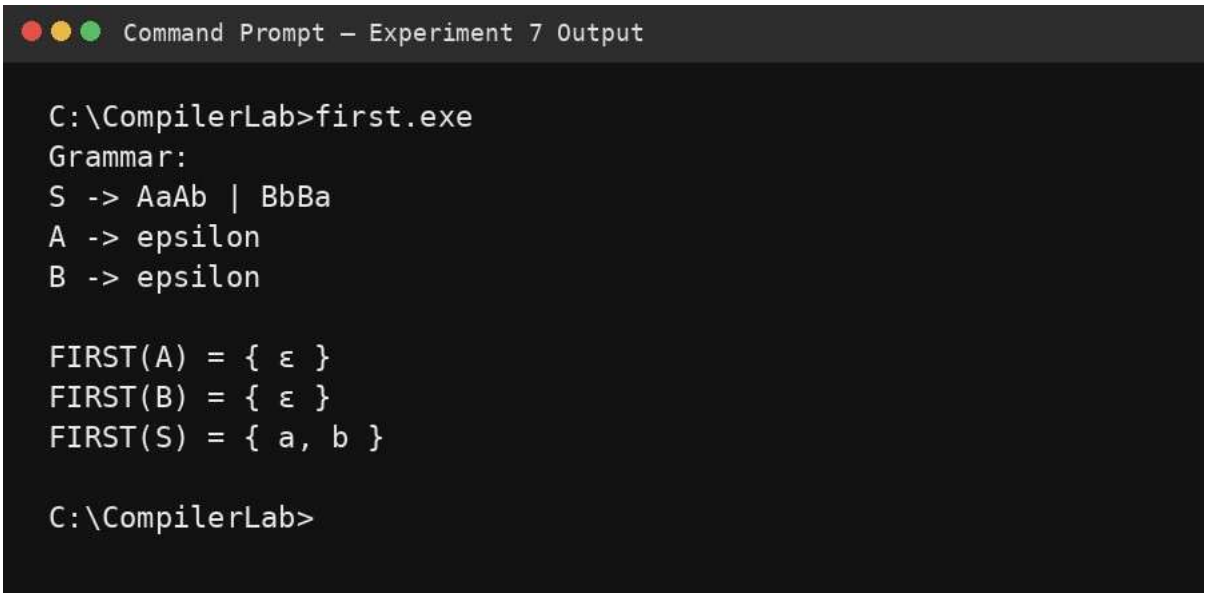
ALGORITHM

- Initialize FIRST of all non-terminals as empty.
- For A → ε, add ε to FIRST(A).
- For B → ε, add ε to FIRST(B).
- For S → AaAb, because A can derive ε, propagate the first terminal a and ε when the complete RHS can derive ε.
- For S → BbBa, similarly propagate b and ε.
- Repeat until no set changes.

PROGRAM

```
#include <stdio.h>

int main(void) {
printf("Grammar:\n");
   printf("S -> AaAb | BbBa\nA -> epsilon\nB -> epsilon\n\n");
printf("FIRST(A) = { epsilon }\n");     printf("FIRST(B) = {
epsilon }\n");     printf("FIRST(S) = { a, b }\n");     return 0;
}
```

OUTPUT



RESULT

Thus, the FIRST sets for the supplied grammar were obtained successfully.

EXPERIMENT 8: FOLLOW () – predictive parser

To compute FOLLOW sets for the grammar S → AaAb / BbBa, A → ε, B → ε.

AIM

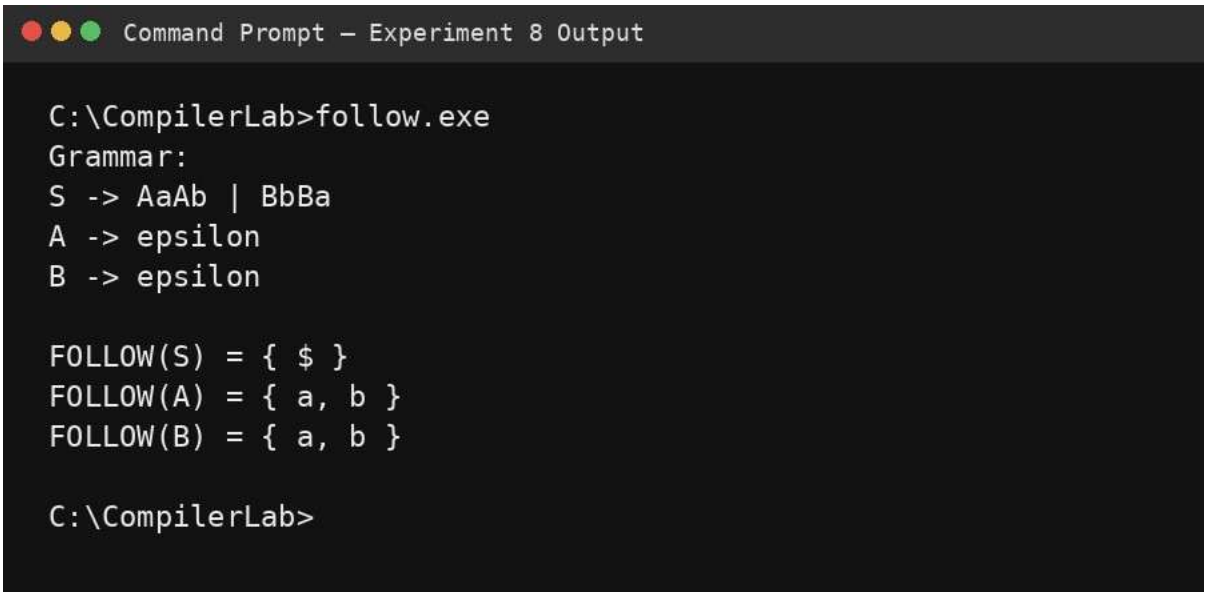
ALGORITHM

- Place $ in FOLLOW(S) because S is the start symbol.
- Scan each production and apply FOLLOW rules to non-terminals.
- For A in S → AaAb, the terminal a follows A, so add a.
- For B in S → BbBa, the terminal b follows B, so add b.
- Propagate the start-symbol marker and any applicable trailer sets until stable.

PROGRAM

```
#include <stdio.h>

int main(void) {
printf("Grammar:\n");
   printf("S -> AaAb | BbBa\nA -> epsilon\nB -> epsilon\n\n");
printf("FOLLOW(S) = { $ }\n");     printf("FOLLOW(A) = {
a, b }\n");
   printf("FOLLOW(B) = { a, b }\n");
return 0; }
```

OUTPUT



RESULT

Thus, the FOLLOW sets for the supplied grammar were obtained successfully.

EXPERIMENT 9: Eliminate left recursion from a CFG

To eliminate immediate left recursion from S → (L) / a and L → L,S / S.

AIM

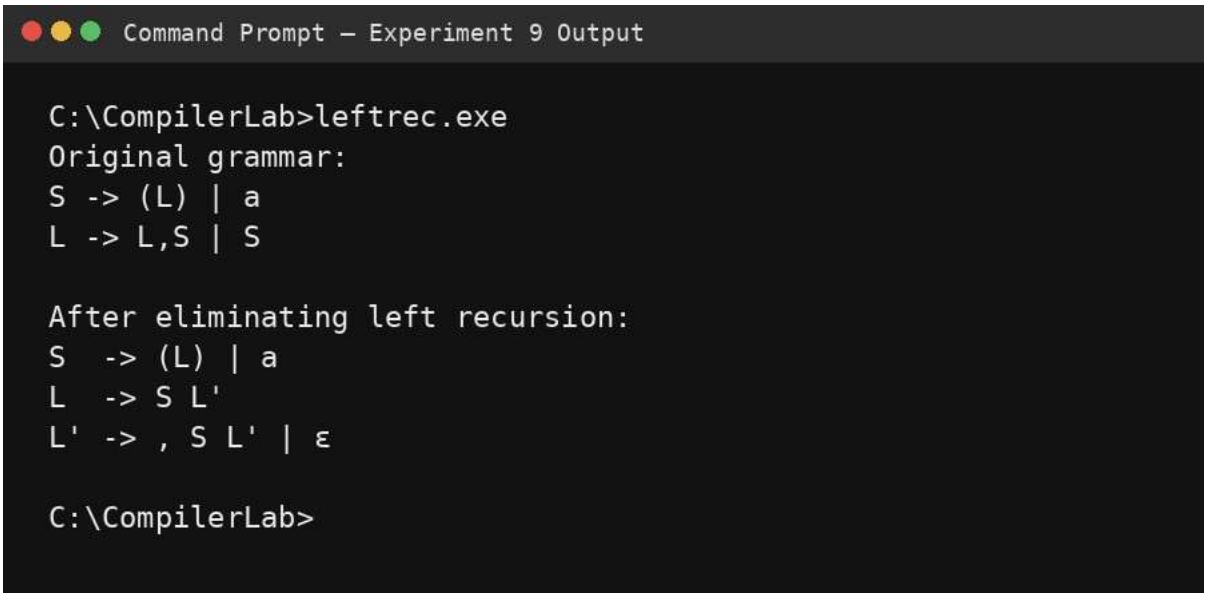
ALGORITHM

- Identify productions of the form A → Aα | β.
- For L, identify α = , and β = S.
- Rewrite L as L → S L'.
- Rewrite L' as L' → , S L' | ε.
- Retain the non-left-recursive production for S.

PROGRAM

```
#include <stdio.h> int
main(void) {
   printf("Original grammar:\n");
printf("S -> (L) | a\n");     printf("L
-> L,S | S\n\n");


   printf("After eliminating left recursion:\n");
   printf("S  -> (L) | a\n");
printf("L  -> S L'\n");     printf("L'
-> , S L' | epsilon\n");     return 0;
}
```

OUTPUT



RESULT

Thus, the immediate left recursion in L was successfully eliminated.

EXPERIMENT 10: Eliminate left factoring from a CFG

To eliminate common prefixes from S → iEtS / iEtSeS / a and E → b.

AIM

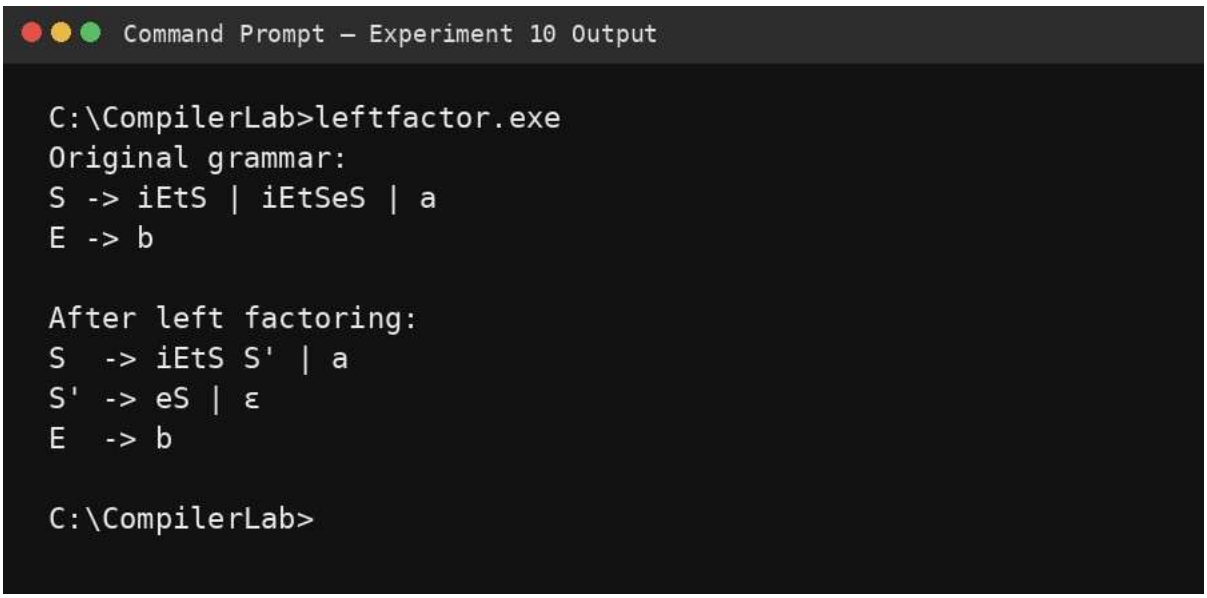
ALGORITHM

- Find the longest common prefix among alternatives of the same non-terminal.
- The first two S productions share the prefix iEtS.
- Factor that prefix as S → iEtS S' | a.
- The remaining alternatives for S' are ε and eS.
- Keep E → b unchanged.

PROGRAM

```
#include <stdio.h> int main(void)
{    printf("Original grammar:\n");
printf("S -> iEtS | iEtSeS | a\n");
printf("E -> b\n\n");

   printf("After left factoring:\n");
printf("S    -> iEtS S' | a\n");
printf("S'   -> eS | epsilon\n");
printf("E  -> b\n");    return 0; }
```

OUTPUT



RESULT

Thus, the common prefix was successfully factored from the grammar.

EXPERIMENT 11: Symbol table operations

To implement insertion, lookup and display operations on a simple compiler symbol table.

AIM

ALGORITHM

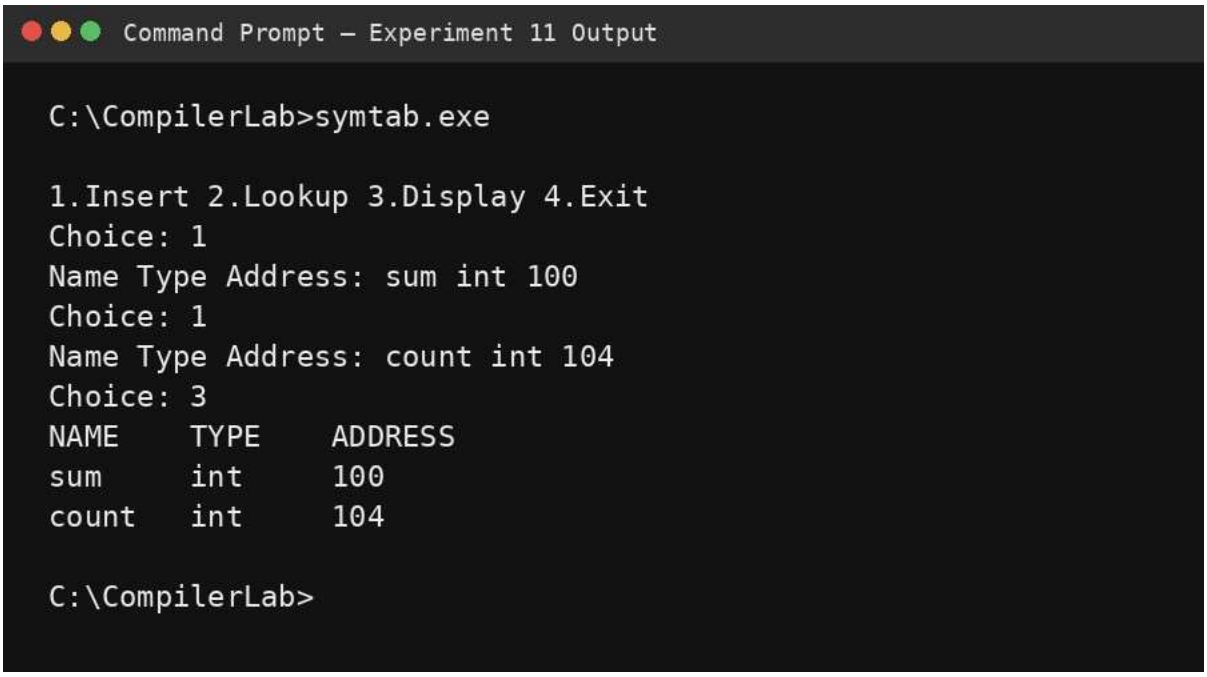
- Create an array of symbol-table entries.
- For insertion, check for duplicates and store a new symbol.
- For lookup, compare the requested name with each stored symbol.
- For display, print all stored entries.
- Demonstrate the operations through a menu.

PROGRAM

```
#include <stdio.h>
#include <string.h>

#define MAX 50
typedef struct { char name[30]; char type[20]; int addr; } Symbol;
Symbol st[MAX];
int n = 0;

void insert(void) {
   char name[30], type[20];
   int i;
   printf("Name Type Address: ");
scanf("%29s %19s %d", name, type, &i);
   for (int k=0;k<n;k++)
      if (!strcmp(st[k].name,name)) { printf("Duplicate symbol\n"); return; }
strcpy(st[n].name,name); strcpy(st[n].type,type);     st[n].addr=i; n++;
}
void lookup(void) {
   char name[30]; printf("Lookup name: "); scanf("%29s",name);
for (int i=0;i<n;i++)         if (!strcmp(st[i].name,name)) {
          printf("Found: %s %s %d\n",st[i].name,st[i].type,st[i].addr);
return;
       }
   printf("Not found\n");
}
void display(void) {
   printf("NAME\tTYPE\tADDRESS\n");
   for (int i=0;i<n;i++) printf("%s\t%s\t%d\n",st[i].name,st[i].type,st[i].addr);
} int main(void)
{     int ch;     do
{
      printf("\n1.Insert 2.Lookup 3.Display 4.Exit\nChoice: ");
scanf("%d",&ch);
      if(ch==1) insert(); else if(ch==2) lookup(); else if(ch==3) display();
   } while(ch!=4);
return 0; }
```

‘

OUTPUT



RESULT

Thus, the basic symbol table insertion, lookup and display operations were implemented successfully.

EXPERIMENT 12: Recursive descent parsing

To construct a recursive descent parser for E → TE', E' → +TE' | ε, T → FT', T' → *FT' | ε, F → (E) | id.

ALGORITHM

- Create one parsing function for each non-terminal.
- Implement E as T followed by E'.
- Implement E' as +TE' or ε.
- Implement T as F followed by T'.
- Implement T' as *FT' or ε.
- Implement F as (E) or id.
- Accept only when the complete input is consumed.

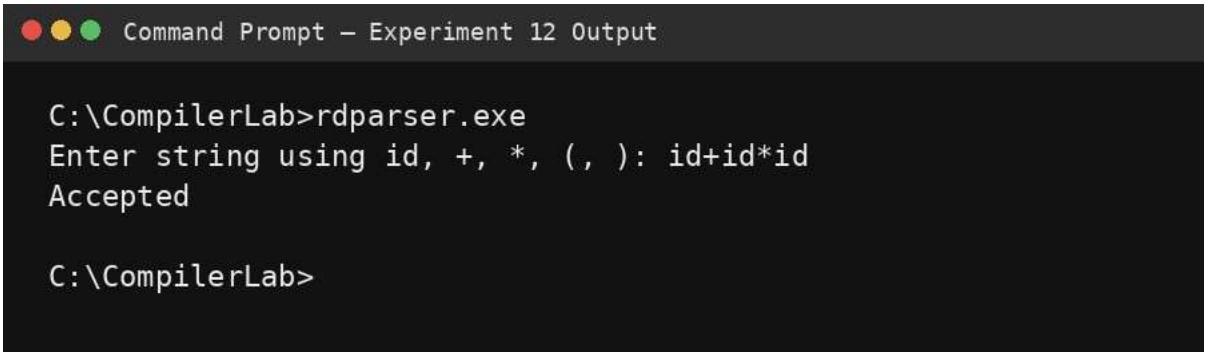
PROGRAM

```
#include <stdio.h>
#include <string.h>

char s[100]; int i=0;
int E(void); int T(void); int F(void);
```

AIM

```
int F(void) {
   if (s[i]=='i' && s[i+1]=='d') { i+=2; return 1; }     if
(s[i]=='(') { i++; if(E() && s[i]==')'){i++; return 1;} }
return 0;
} int T(void) {
if(!F()) return 0;
   while(s[i]=='*') { i++; if(!F()) return 0; }
return 1;
} int E(void) {
if(!T()) return 0;
   while(s[i]=='+') { i++; if(!T()) return 0; }
return 1;
}
int main(void) {
   printf("Enter string using id, +, *, (, ): ");
   scanf("%99s",s);
   printf(E() && s[i]=='\0' ? "Accepted\n" : "Rejected\n");
return 0; }
```

OUTPUT



RESULT

Thus, a recursive descent parser for the supplied expression grammar was implemented successfully.

AIM

EXPERIMENT 13: Top-down parsing to validate a grammar

To use a top-down parsing technique to determine whether an input string satisfies a context-free grammar.

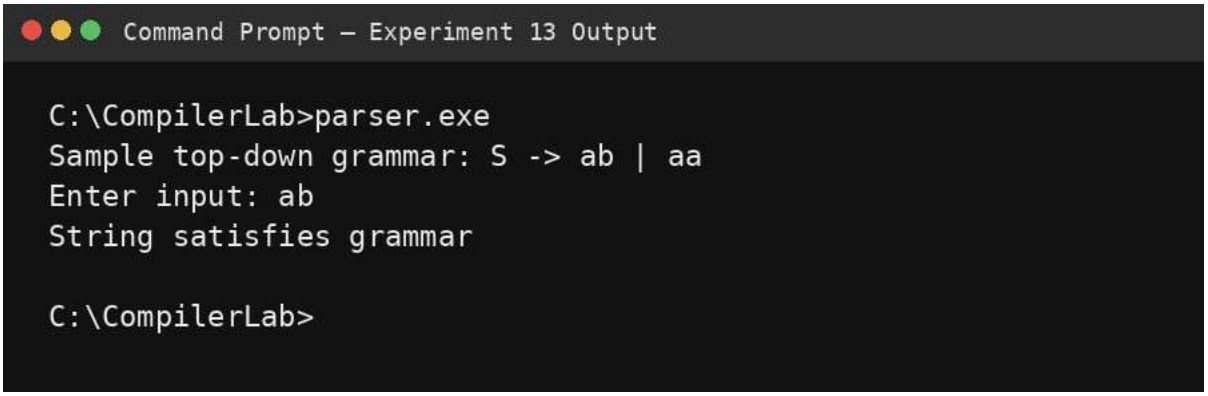
ALGORITHM

- Select the supplied grammar and identify the start symbol.
- Implement procedures corresponding to non-terminals.
- Expand productions from the start symbol toward terminals.
- Match each terminal against the input.
- Accept when the entire input is consumed with no unmatched symbols.
- Otherwise report rejection.

PROGRAM

```
#include <stdio.h>
#include <string.h>

char s[100]; int i=0;
int S(void) {     int
save=i;
   if(s[i]=='a'){ i++; if(s[i]=='b'){i++; return 1;} }
i=save;
   if(s[i]=='a'){ i++; if(s[i]=='a'){i++; return 1;} }
return 0;
}
int main(void){
   printf("Sample top-down grammar: S -> ab | aa\n");
   printf("Enter input: "); scanf("%99s",s);
i=0;
   printf(S() && s[i]=='\0' ? "String satisfies grammar\n"
: "String does not satisfy grammar\n");     return 0;
}
```

OUTPUT



RESULT

Thus, top-down parsing was used to validate an input string against a context-free grammar.

EXPERIMENT 14: Three-address code generation

AIM

To generate a three-address-code representation for an input arithmetic expression.

ALGORITHM

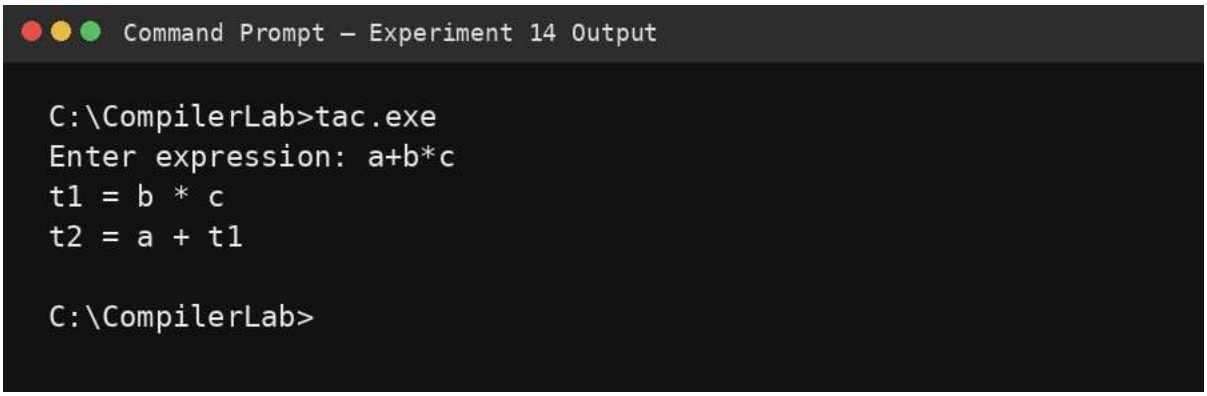
- Read an infix expression.
- Use operator precedence to convert it to postfix notation.
- For every operator, create a temporary variable.
- Emit a three-address statement using the operands and temporary.
- Continue until the final temporary represents the complete expression.

PROGRAM

```
#include <stdio.h>
#include <ctype.h>
#include <string.h>

int prec(char c){ return c=='+'||c=='-'?1:c=='*'||c=='/'?2:0; }

int main(void){     char in[100], st[100], post[100];
int top=-1, k=0;     printf("Enter expression: ");
scanf("%99s",in);

   for(int i=0; in[i]; i++){
      if(isalnum((unsigned char)in[i])) post[k++]=in[i];
      else if(in[i]=='(') st[++top]=in[i];
      else if(in[i]==')'){
         while(top>=0 && st[top]!='(') post[k++]=st[top--];
if(top>=0) top--;
      } else {
         while(top>=0 && prec(st[top])>=prec(in[i])) post[k++]=st[top--];
st[++top]=in[i];
      }
   }
   while(top>=0) post[k++]=st[top--];
   post[k]='\0';

   char a[20][10]; int n=0, t=1;     for(int
i=0; post[i]; i++){
if(isalnum((unsigned char)post[i])) {
a[n][0]=post[i]; a[n][1]='\0'; n++;
      } else {
         char temp[10], x[10], y[10];
strcpy(y,a[--n]); strcpy(x,a[--n]);
sprintf(temp,"t%d",t++);            printf("%s = %s
%c %s\n",temp,x,post[i],y);
         strcpy(a[n++],temp);
      }     }
return 0;
}
```

AIM

OUTPUT



RESULT

Thus, the given arithmetic expression was represented successfully in three-address code.

EXPERIMENT 15: Lexical Analyzer – count characters, words and lines in a file

AIM

To scan a file and count its characters, words and lines using a C program.

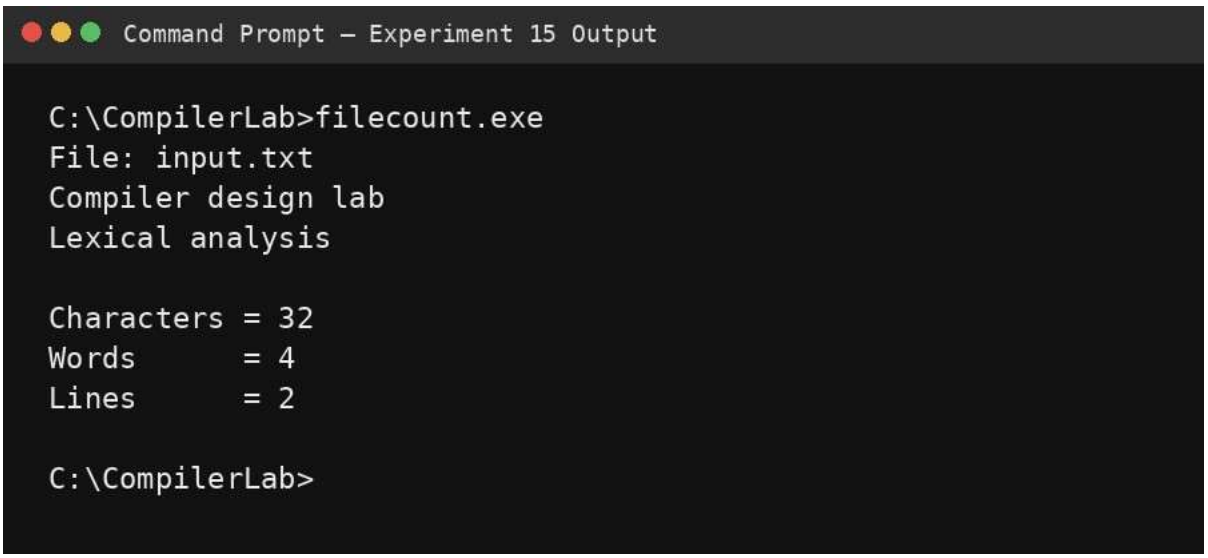
ALGORITHM

- Open the input file in read mode.
- Read characters until EOF.
- Increment the character count for each character.
- Use white space boundaries to detect words.
- Increment the line count whenever a newline is encountered.
- Display the totals.

PROGRAM

```
#include <stdio.h>
#include <ctype.h> int
main(void) {
   FILE *fp = fopen("input.txt","r");     int c,
inword=0, chars=0, words=0, lines=0;
if(!fp){ perror("input.txt"); return 1; }
while((c=fgetc(fp))!=EOF){
      chars++;        if(c=='\n')
lines++;        if(isspace(c))
inword=0;
      else if(!inword){ words++; inword=1; }
   }
fclose(fp);
   printf("Characters = %d\nWords      = %d\nLines      = %d\n",chars,words,lines);
return 0;
}
```

OUTPUT



RESULT

Thus, the file was scanned and the number of characters, words and lines was obtained successfully.

EXPERIMENT 16: Back end of the compiler

AIM

To demonstrate a simple compiler back-end step that converts intermediate three-address statements into target-like instructions.

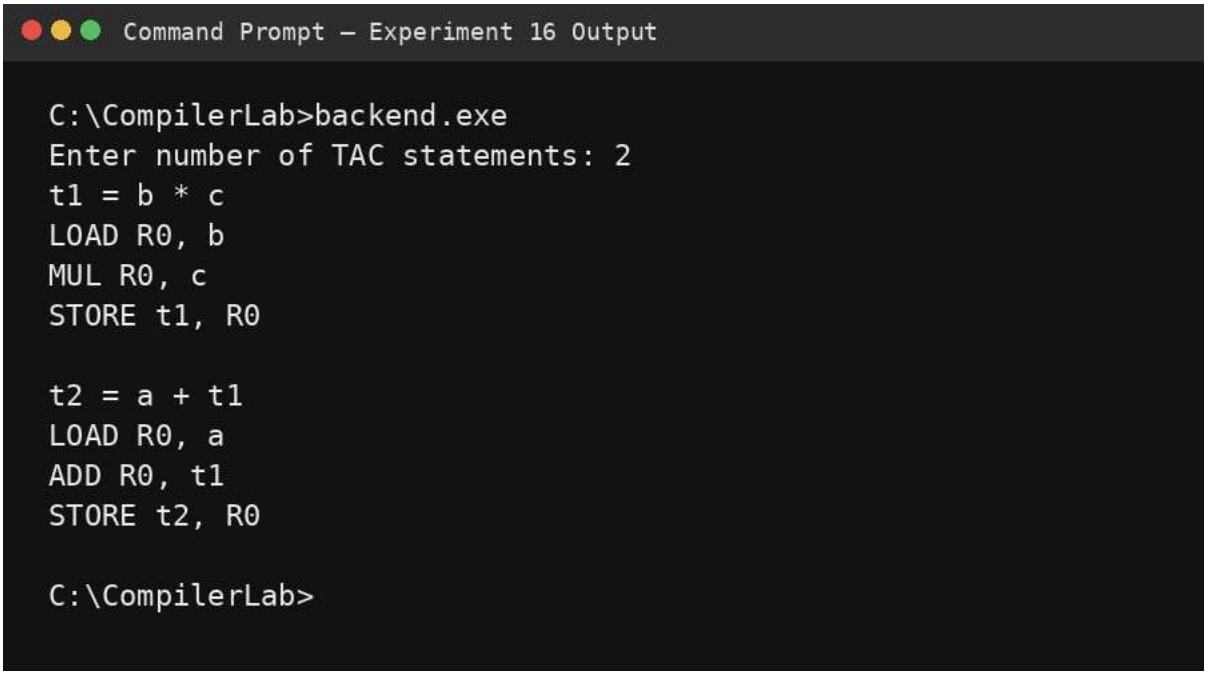
ALGORITHM

- Read each three-address statement of the form t = x op y.
- Load the first operand into a register.
- Generate the target instruction for the operator.
- Store the result into the target location.
- Repeat for all intermediate statements.

PROGRAM

```
#include <stdio.h>

int main(void){
   char t[20], x[20], op, y[20];
   int n;
   printf("Enter number of TAC statements: ");
scanf("%d",&n);     while(n--){
      scanf("%19s = %19s %c %19s",t,x,&op,y);
printf("LOAD R0, %s\n",x);
      switch(op){           case '+': printf("ADD
R0, %s\n",y); break;          case '-': printf("SUB
R0, %s\n",y); break;          case '*': printf("MUL
R0, %s\n",y); break;          case '/': printf("DIV
R0, %s\n",y); break;
      }
      printf("STORE %s, R0\n\n",t);
   }
return 0; }
```

OUTPUT



RESULT

Thus, a simple illustrative back-end translation from three-address code to target-like instructions was implemented successfully.

AIM

EXPERIMENT 17: LEADING( ) – operator precedence parser

To compute LEADING sets for E → E+T | T, T → T*F | F, F → (E) | id.

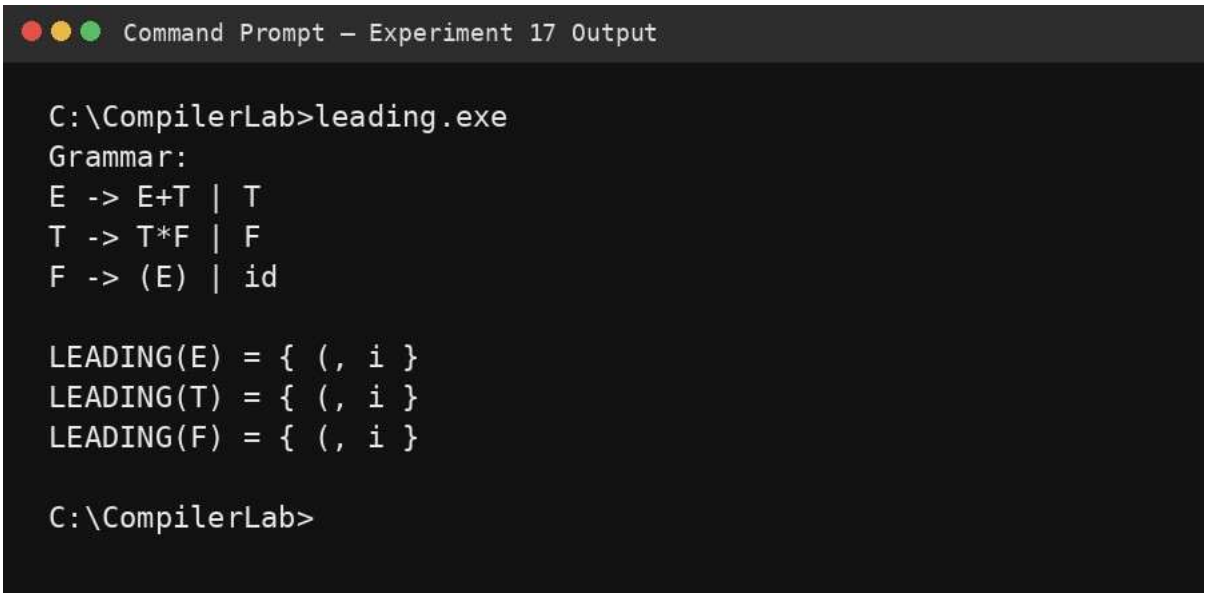
ALGORITHM

- Initialize LEADING sets as empty.
- For a production beginning with a terminal, add that terminal.
- For a production beginning with a non-terminal, copy its LEADING set and account for the following terminal where applicable.
- Repeat until no set changes.
- Display the final LEADING sets.

PROGRAM

```
#include <stdio.h>

int main(void){
   printf("Grammar:\nE -> E+T | T\nT -> T*F | F\nF -> (E) | id\n\n");
printf("LEADING(E) = { (, i }\n");     printf("LEADING(T) = { (, i
}\n");     printf("LEADING(F) = { (, i }\n");     return 0;
}
```

OUTPUT



RESULT

Thus, the LEADING sets for the supplied operator-precedence grammar were obtained successfully.

AIM

EXPERIMENT 18: TRAILING( ) – operator precedence parser

To compute TRAILING sets for E → E+T | T, T → T*F | F, F → (E) | id.

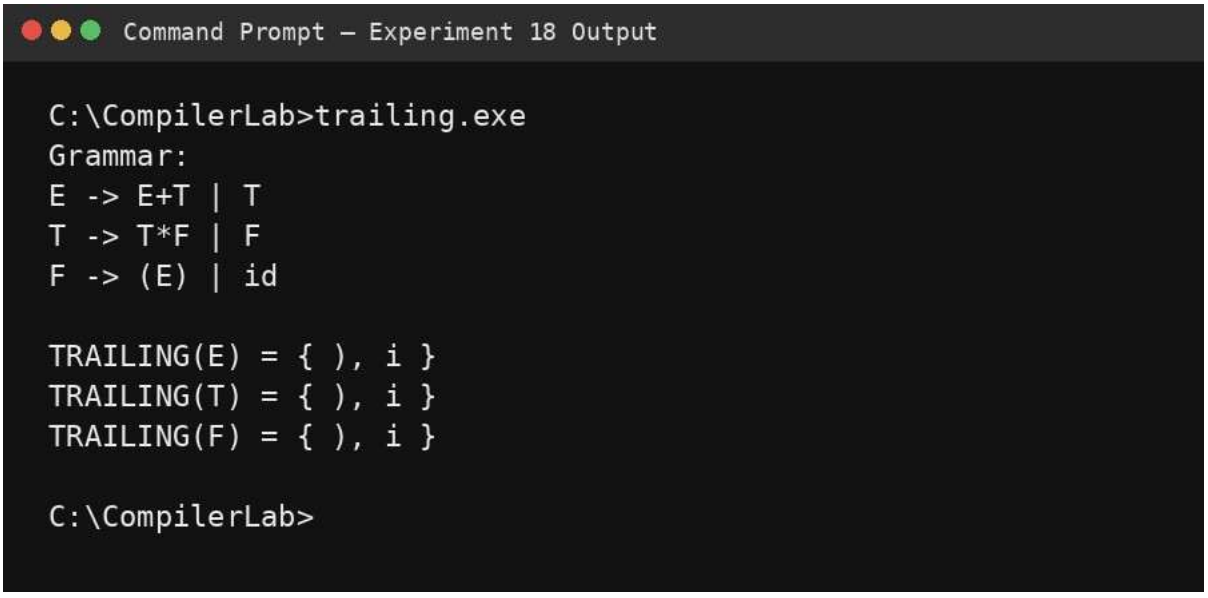
ALGORITHM

- Initialize TRAILING sets as empty.
- For a production ending in a terminal, add that terminal.
- For a production ending in a non-terminal, propagate its TRAILING set and account for preceding terminals where applicable.
- Repeat until stable.
- Display the final sets.

PROGRAM

```
#include <stdio.h>

int main(void){
   printf("Grammar:\nE -> E+T | T\nT -> T*F | F\nF -> (E) | id\n\n");
printf("TRAILING(E) = { ), i }\n");    printf("TRAILING(T) = { ), i
}\n");    printf("TRAILING(F) = { ), i }\n");    return 0;
}
```

OUTPUT



RESULT

Thus, the TRAILING sets for the supplied grammar were obtained successfully.

AIM

EXPERIMENT 19: LEX – count characters, lines and words in a C file

To write a LEX specification that scans a C source file and counts characters, lines and words.

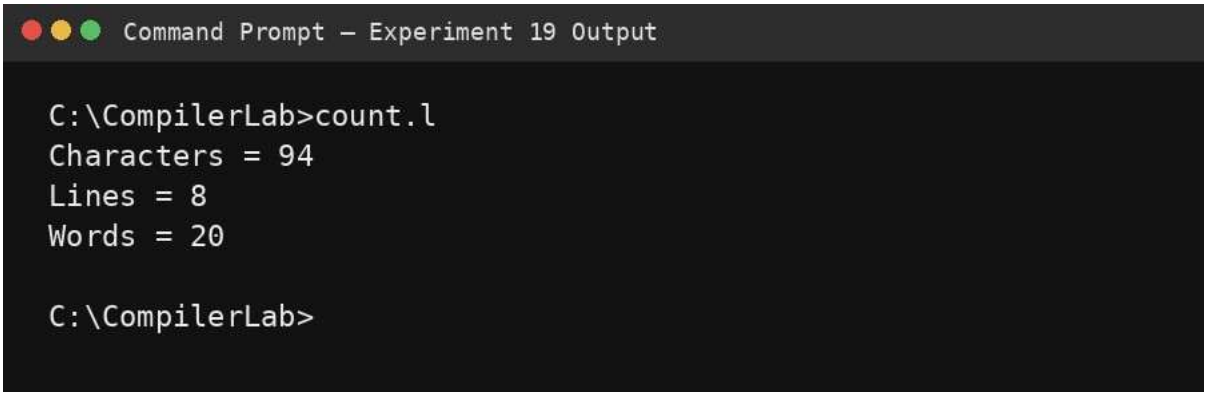
ALGORITHM

- Define patterns for newlines, white space, words and any other character.
- Update counters in the corresponding LEX actions.
- Ignore or count comments according to the required scanner policy.
- Run the scanner on the C source file.
- Print the final counts.

PROGRAM

```
%{
#include <stdio.h>
int chars=0, words=0, lines=0;
%}
%%
\n              { chars++; lines++; }
[ \t]+          { chars += yyleng; }
[A-Za-z_][A-Za-z0-9_]* { chars += yyleng; words++; }
[0-9]+          { chars += yyleng; words++; }
.               { chars++; }
%%
int yywrap(void){ return 1; }
int main(void){     yylex();
   printf("Characters = %d\nLines = %d\nWords = %d\n",chars,lines,words);
return 0;
}
```

OUTPUT



RESULT

Thus, the LEX specification successfully scanned the supplied C source and counted characters, lines and words.

AIM

EXPERIMENT 20: LEX – print constants in a C source file

To print all numeric constants occurring in a C source program.

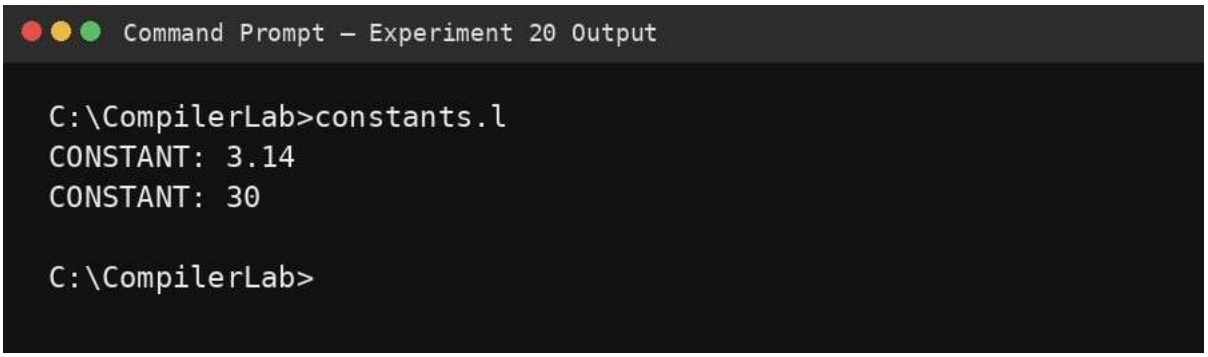
ALGORITHM

- Define a LEX rule for integer and floating-point constants.
- Ignore identifiers and other source characters.
- Print every matched constant.
- Run the scanner over sample.c.

PROGRAM

```
%{
#include <stdio.h>
%}
%%
[0-9]+(\.[0-9]+)?    { printf("CONSTANT: %s\n", yytext); }
[ \t\n]+          ;
.                 ;
%%
int yywrap(void){ return 1; } int
main(void){ yylex(); return 0; }
```

OUTPUT



RESULT

Thus, the numeric constants in the supplied sample C source were identified and printed successfully.

EXPERIMENT 21: Algorithm – count vowels in a sentence

To count the number of vowels in a given sentence.

AIM

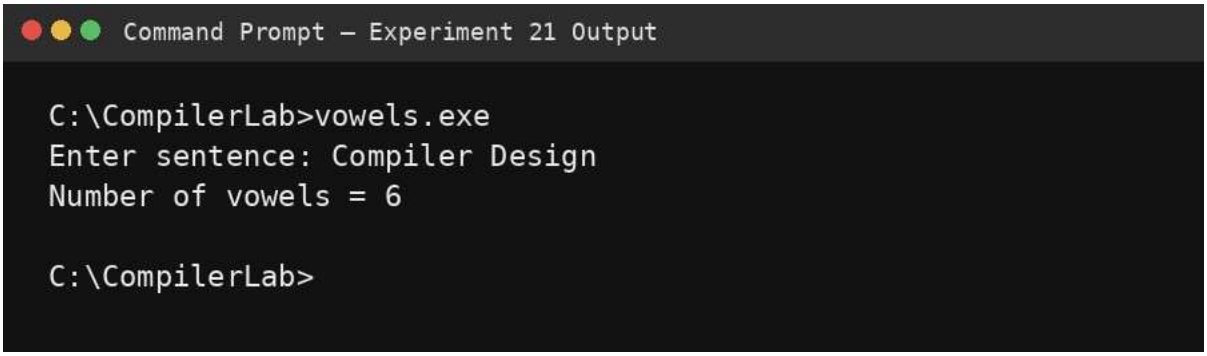
ALGORITHM

- Read the sentence.
- Initialize the vowel count to zero.
- For every character, convert it conceptually to lower case.
- If the character is a, e, i, o or u, increment the count.
- Display the total number of vowels.

PROGRAM

```
#include <stdio.h>
#include <ctype.h>

int main(void){     char s[500]; int count=0;
printf("Enter sentence: ");
fgets(s,sizeof(s),stdin);     for(int i=0;s[i];i++){
char c=(char)tolower((unsigned char)s[i]);
if(c=='a'||c=='e'||c=='i'||c=='o'||c=='u') count++;
   }
   printf("Number of vowels = %d\n",count);
return 0;
}
```

OUTPUT



RESULT

Thus, the number of vowels in the given sentence was counted successfully.

EXPERIMENT 22: LEX – print HTML tags

To write a LEX program that prints all HTML tags in an input file.

ALGORITHM

- Define a pattern beginning with < and ending with >.
- Print each matched tag.

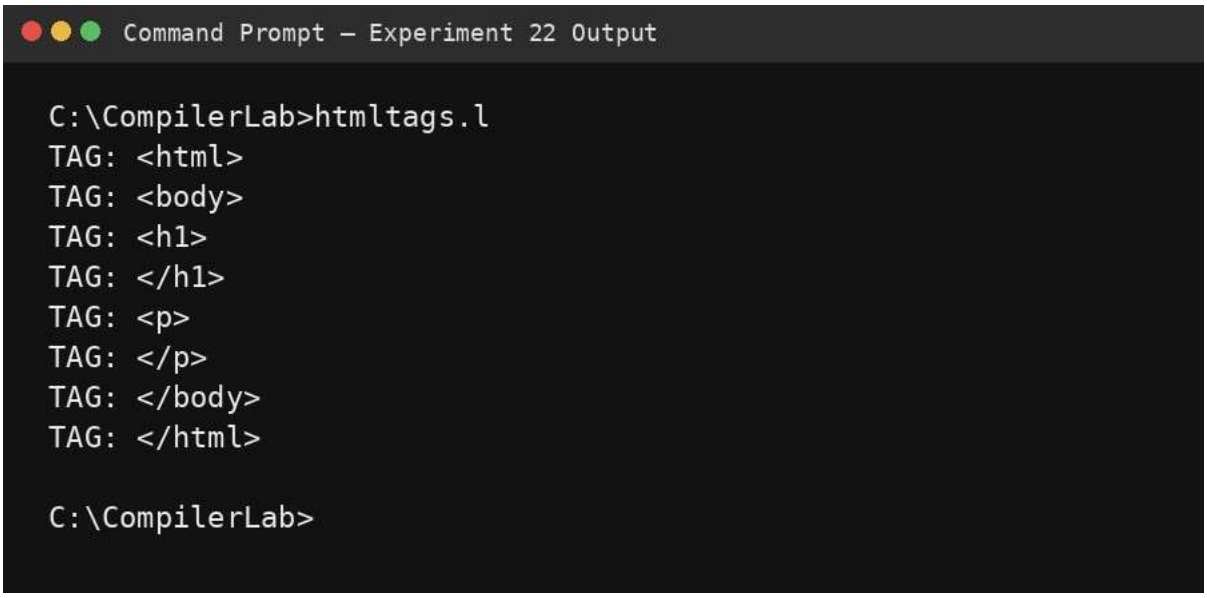
AIM

- Ignore text outside tags.
- Run the scanner on sample.html.

PROGRAM

```
%{
#include <stdio.h>
%}
%%
<[^>]+>     { printf("TAG: %s\n", yytext); }
\n          ;
.           ;
%%
int yywrap(void){ return 1; } int
main(void){ yylex(); return 0; }
```

OUTPUT



RESULT

Thus, all HTML tags in the supplied sample input were printed successfully.

EXPERIMENT 23: LEX – add line numbers to a C program

To add sequential line numbers to every line of a C source file and display the numbered program.

ALGORITHM

- Match a newline and increment the line counter.
- At the beginning of each line, print the current line number.

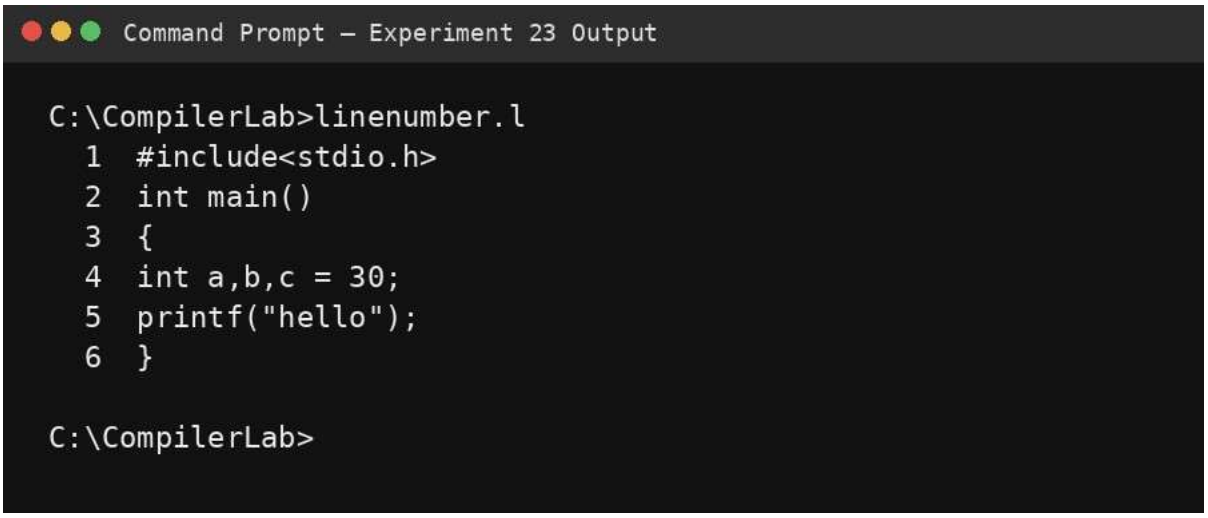
AIM

- Print all characters of the line.
- Continue until EOF.

PROGRAM

```
%{
#include <stdio.h>
int line=1, at_start=1;
%}
%%
^[^\n]*      { printf("%3d  %s\n",line++,yytext); }
\n           ;
%%
int yywrap(void){ return 1; } int
main(void){ yylex(); return 0; }
```

OUTPUT



RESULT

Thus, line numbers were added and the numbered source was displayed successfully.

EXPERIMENT 24: LEX – count and remove comments from a C program

AIM

To count comment lines in a C program, remove comments, and write the remaining source into another file.

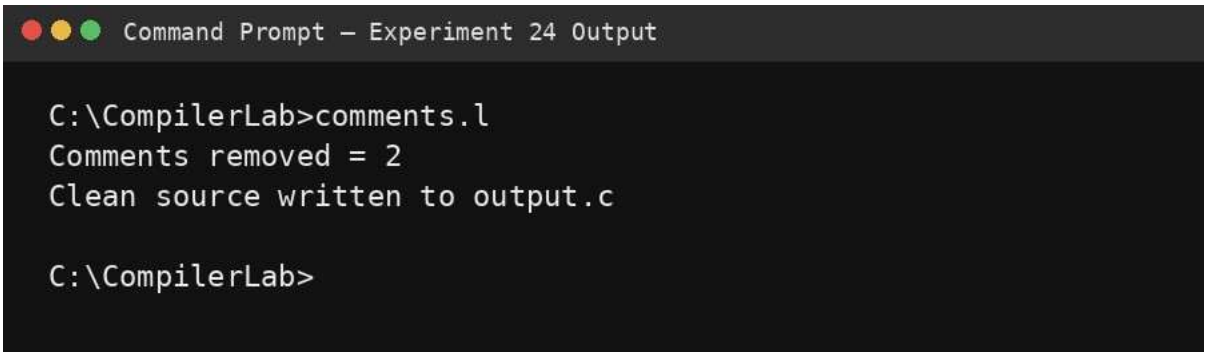
ALGORITHM

- Open the input and output streams.
- Match // comments and /*...*/ comments.
- Increment the comment counter for each matched comment.
- Write non-comment text to the output file.
- Display the number of comments removed.

PROGRAM

```
%{
#include <stdio.h> int
comments=0;
FILE *out;
%}
%x BLOCK
%%
"//"[^\n]*       { comments++; }
"/*"             { comments++; BEGIN(BLOCK); }
<BLOCK>"*/"      { BEGIN(INITIAL); }
<BLOCK>\n        ;
<BLOCK>.         ;
\n               { fputc('\n',out); }
.                { fputc(yytext[0],out); }
%%
int yywrap(void){ return 1; } int
main(void){
   FILE *in=fopen("input.c","r");
   out=fopen("output.c","w");
if(!in   ||   !out)   return   1;
yyin=in; yylex();
   fclose(in); fclose(out);
   printf("Comments removed = %d\n",comments);
printf("Clean source written to output.c\n");     return
0;
}
```

OUTPUT



RESULT

AIM

Thus, comments were counted and removed, and the cleaned source was written to the output file.

EXPERIMENT 25: LEX – identify capital words

To identify and print words that are written entirely in capital letters.

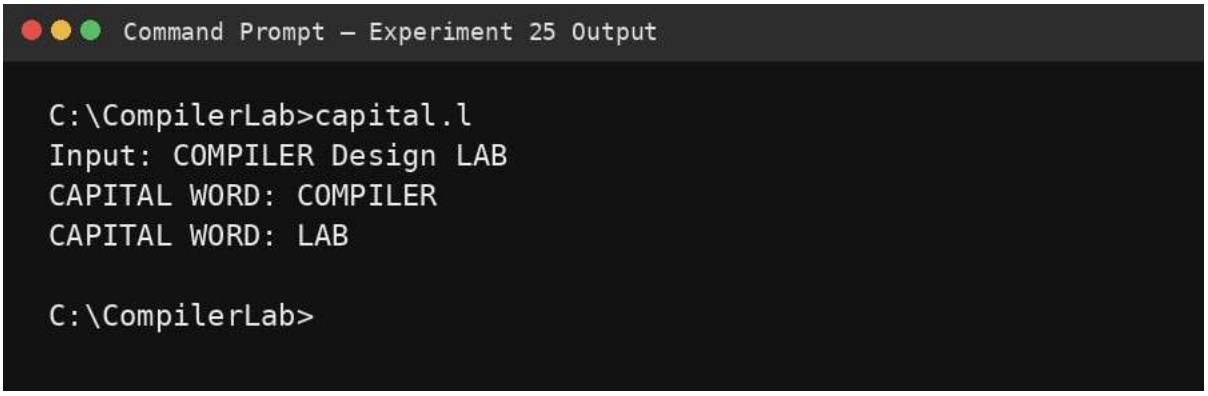
ALGORITHM

- Define a word pattern containing only uppercase alphabetic characters.
- Print each matched word.
- Ignore lower-case words and other characters.

PROGRAM

```
%{
#include <stdio.h>
%}
%%
[A-Z]+      { printf("CAPITAL WORD: %s\n", yytext); }
[A-Za-z]+   ;
[ \t\n]+    ;
.           ;
%%
int yywrap(void){ return 1; } int
main(void){ yylex(); return 0; }
```

OUTPUT



RESULT

Thus, the capital words in the input were identified successfully.

AIM

EXPERIMENT 26: LEX validate an email address

To write a LEX program that checks whether an input string follows a basic email-address pattern.

ALGORITHM

- Define a local-part pattern using letters, digits and common punctuation.
- Require the @ symbol.
- Require a domain and a dot-separated extension.
- Match the complete input and report valid or invalid.

PROGRAM

```
%{
#include <stdio.h>
%}
%%
^[A-Za-z0-9._%+-]+@[A-Za-z0-9.-]+\.[A-Za-z]{2,}$  { printf("Valid email address\n"); }
.*$                                    { printf("Invalid email address\n"); }
%%
int yywrap(void){ return 1; } int
main(void){ yylex(); return 0; }
```

OUTPUT

RESULT

Thus, the supplied email address was validated using a LEX regular expression.

EXPERIMENT 27: LEX – convert abc to ABC

To replace every occurrence of the substring abc with ABC in the input.

AIM

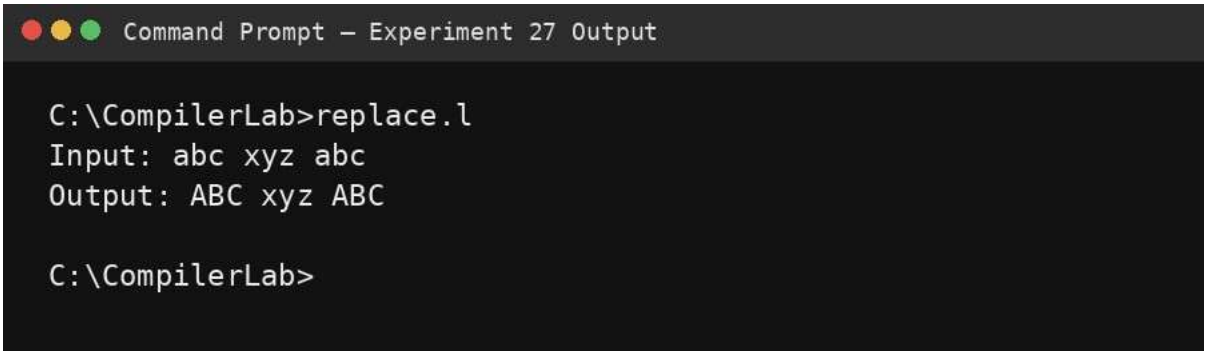
ALGORITHM

- Match the exact lowercase substring abc.
- Print ABC whenever it is matched.
- Print all other characters unchanged.

PROGRAM

```
%{
#include <stdio.h>
%} %%
abc         { printf("ABC"); }
.           { ECHO; }
\n          { ECHO; }
%%
int yywrap(void){ return 1; } int
main(void){ yylex(); return 0; }
```

OUTPUT



RESULT

Thus, every occurrence of the substring abc was converted to ABC successfully.

EXPERIMENT 28: LEX validate mobile numbers

To validate mobile numbers using a LEX pattern.

ALGORITHM

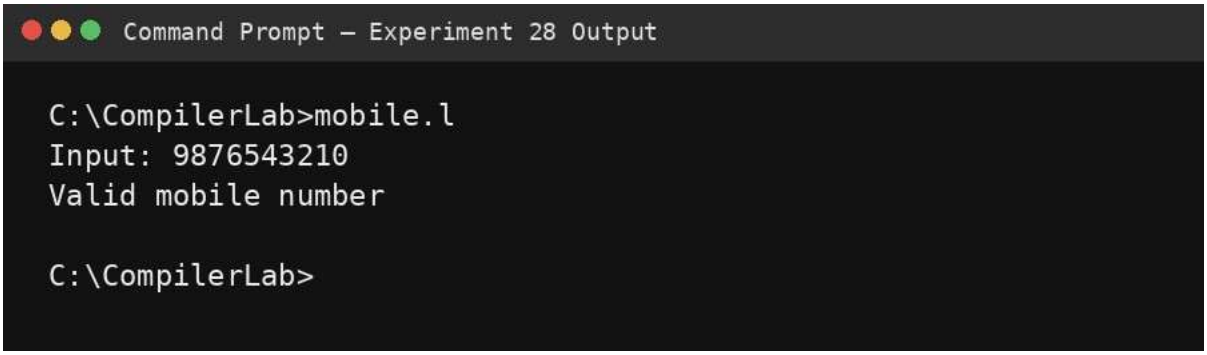
- Read the complete input number.
- Accept a ten-digit number beginning with 6, 7, 8 or 9.
- Reject any other format.
- Display the validation result.

AIM

PROGRAM

```
%{
#include <stdio.h>
%}
%%
^[6-9][0-9]{9}$   { printf("Valid mobile number\n"); }
.*$               { printf("Invalid mobile number\n"); }
%%
int yywrap(void){ return 1; } int
main(void){ yylex(); return 0; }
```

OUTPUT



RESULT

Thus, the mobile number was validated according to the ten-digit laboratory pattern.

AIM

EXPERIMENT 29: LEX/FLEX separate C tokens with captions

To separate tokens in a C program and display each token with an appropriate caption.

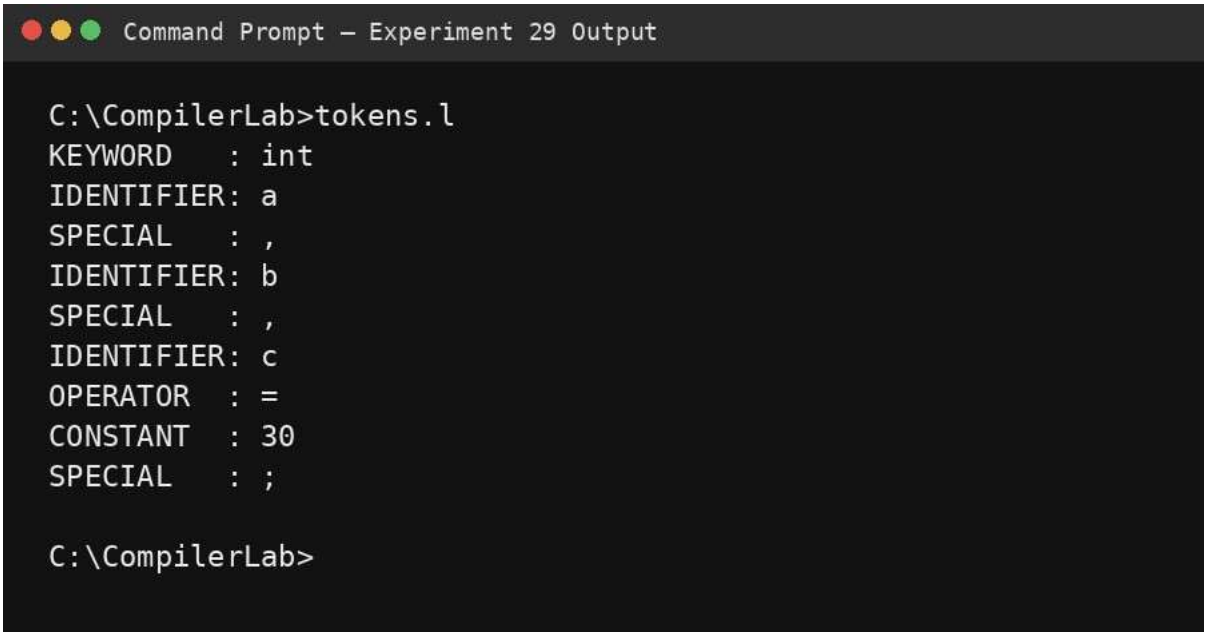
ALGORITHM

- Define patterns for keywords, identifiers, constants, strings, operators and punctuation.
- Scan the C source file from left to right.
- Print each matched lexeme with its token class.
- Ignore comments and white space.

PROGRAM

```
%{
#include <stdio.h>
%}
%%
"int"|"float"|"char"|"return"|"void"   { printf("KEYWORD   : %s\n",yytext); }
[A-Za-z_][A-Za-z0-9_]*                 { printf("IDENTIFIER: %s\n",yytext); }
[0-9]+(\.[0-9]+)?                      { printf("CONSTANT  : %s\n",yytext); }
\"([^\\\"]|\\.)*\"                     { printf("STRING    : %s\n",yytext); }
"=="|"!="|"<="|">="|"+"|"-"|"*"|"/"|"=" { printf("OPERATOR  : %s\n",yytext); }
[(){};,]                               { printf("SPECIAL   : %s\n",yytext); }
[ \t\n]+                               ;
.                                      ;
%%
int yywrap(void){ return 1; } int
main(void){ yylex(); return 0; }
```

OUTPUT



AIM

RESULT

Thus, the C source tokens were separated and displayed with appropriate captions.

EXPERIMENT 30: Algorithm count consonants in a sentence

To count consonants in a given word or sentence.

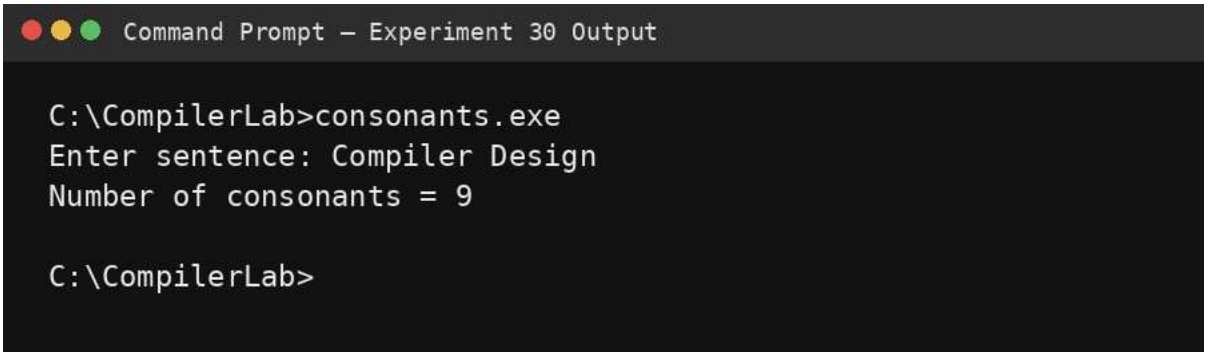
ALGORITHM

- Read the sentence.
- For every alphabetic character, convert it to lower case.
- If it is not a, e, i, o or u, increment the consonant count.
- Ignore spaces, digits and punctuation.
- Display the total consonant count.

PROGRAM

```
#include <stdio.h>
#include <ctype.h>

int main(void){     char
s[500]; int count=0;
printf("Enter sentence: ");
fgets(s,sizeof(s),stdin);
for(int i=0;s[i];i++){
       char c=(char)tolower((unsigned char)s[i]);
if(isalpha((unsigned char)c) &&
          !(c=='a'||c=='e'||c=='i'||c=='o'||c=='u')) count++;
   }
   printf("Number of consonants = %d\n",count);
return 0;
}
```

OUTPUT



AIM

RESULT

Thus, the consonants in the given sentence were counted successfully.

EXPERIMENT 31: LEX – separate keywords and identifiers

To separate reserved keywords from identifiers in a C source program using LEX.

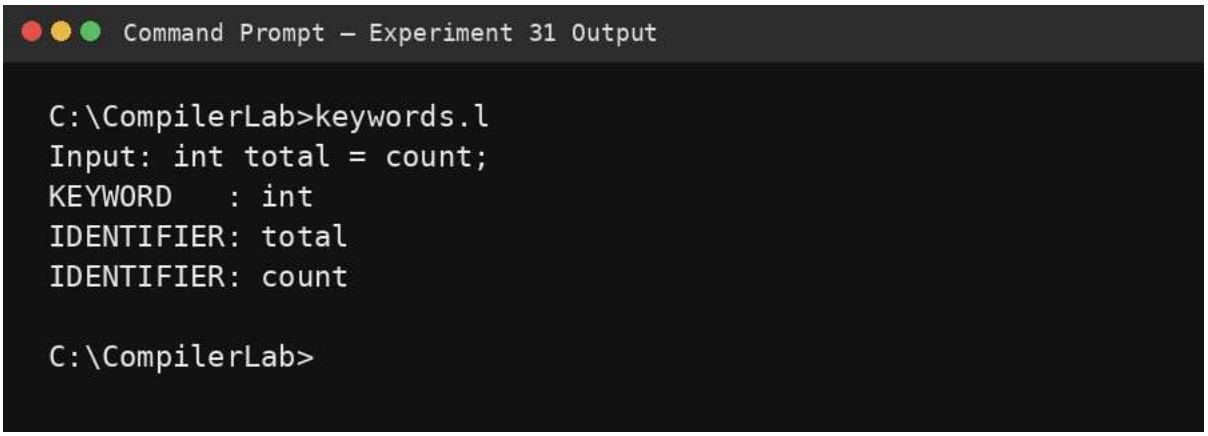
ALGORITHM

- Define the required C keywords as a higher-priority rule.
- Define the general identifier rule after the keyword rule.
- Print the matching class for each lexeme.
- Ignore spaces and other characters.

PROGRAM

```
%{
#include <stdio.h>
%}
%%
"auto"|"break"|"case"|"char"|"const"|"continue"|"default"|"do"|
"double"|"else"|"enum"|"extern"|"float"|"for"|"goto"|"if"|
"int"|"long"|"register"|"return"|"short"|"signed"|"sizeof"|
"static"|"struct"|"switch"|"typedef"|"union"|"unsigned"|
"void"|"volatile"|"while"   { printf("KEYWORD   : %s\n",yytext); }
[A-Za-z_][A-Za-z0-9_]*      { printf("IDENTIFIER: %s\n",yytext); }
[ \t\n]+                    ;
.                           ;
%%
int yywrap(void){ return 1; } int
main(void){ yylex(); return 0; }
```

OUTPUT



AIM

RESULT

Thus, keywords and identifiers were separated successfully using LEX.

AIM

EXPERIMENT 32: LEX – count positive and negative numbers

To identify and count positive and negative numeric values in the input.

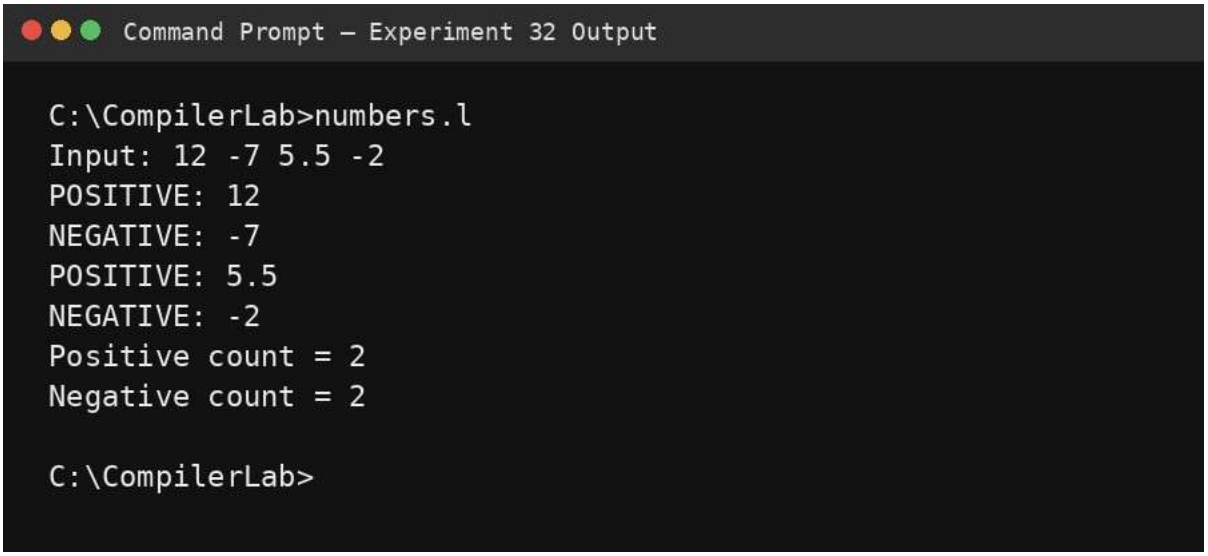
ALGORITHM

- Define a signed-number rule for negative values.
- Define a positive-number rule for numbers without a leading minus.
- Increment the corresponding counter.
- Print each matched number and the final totals.

PROGRAM

```
%{
#include <stdio.h>
int pos=0, neg=0;
%}
%%
-[0-9]+(\.[0-9]+)? { printf("NEGATIVE: %s\n",yytext); neg++; }
\+?[0-9]+(\.[0-9]+)? { printf("POSITIVE: %s\n",yytext); pos++; }
[ \t\n]+ ;
. ;
%%
int yywrap(void){ return 1; }
int main(void){     yylex();
   printf("Positive count = %d\nNegative count = %d\n",pos,neg);
return 0;
}
```

OUTPUT



RESULT

Thus, positive and negative numbers were identified and counted successfully.

AIM

EXPERIMENT 33: LEX – validate a URL

To validate a URL using a LEX regular expression.

ALGORITHM

- Require a protocol such as http or https.
- Require a domain name.
- Allow an optional port and path.
- Match the complete input and report the result.

PROGRAM

```
%{
#include <stdio.h>
%}
%%
^https?://[A-Za-z0-9.-]+\.[A-Za-z]{2,}([/:][^[:space:]]*)?$ { printf("Valid URL\n"); } .*$
{ printf("Invalid URL\n"); }
%%
int yywrap(void){ return 1; } int
main(void){ yylex(); return 0; }
```

OUTPUT

RESULT

Thus, the supplied URL was validated using a laboratory-level LEX pattern.

EXPERIMENT 34: LEX – validate date of birth

To validate a date of birth in the DD/MM/YYYY format.

AIM

ALGORITHM

- Require exactly two digits for day.
- Require exactly two digits for month.
- Require exactly four digits for year.
- Accept the syntactic format DD/MM/YYYY.
- Display valid or invalid.

PROGRAM

```
%{
#include <stdio.h>
%}
%%
^(0[1-9]|[12][0-9]|3[01])/(0[1-9]|1[0-2])/[0-9]{4}$ { printf("Valid DOB format\n"); }
.*$ { printf("Invalid DOB format\n"); }
%%
int yywrap(void){ return 1; } int
main(void){ yylex(); return 0; }
```

OUTPUT

RESULT

Thus, the date was validated for the required DD/MM/YYYY syntactic format.

EXPERIMENT 35: LEX – check whether input is a digit

To determine whether the given input character is a digit.

ALGORITHM

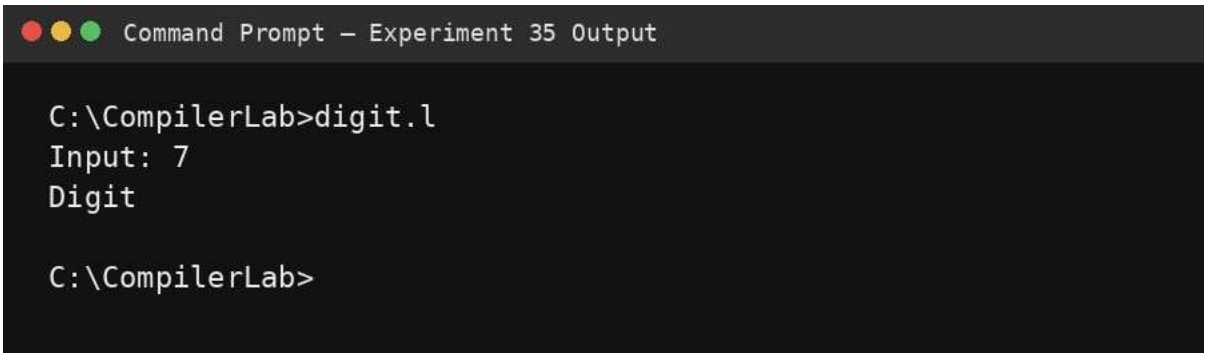
- Read the input character.
- Match it against the range 0–9.
- If it matches, print digit; otherwise print not a digit.

AIM

PROGRAM

```
%{
#include <stdio.h>
%}
%%
^[0-9]$   { printf("Digit\n"); }
^.$       { printf("Not a digit\n"); }
%%
int yywrap(void){ return 1; } int
main(void){ yylex(); return 0; }
```

OUTPUT



RESULT

Thus, the input character was correctly classified as a digit or non-digit.

EXPERIMENT 36: LEX – basic mathematical operations

To recognize basic arithmetic operators and numeric operands using LEX.

ALGORITHM

- Define a numeric operand pattern.
- Define patterns for +, -, *, and /.
- Print the recognized token class for each input lexeme.
- Ignore white space.

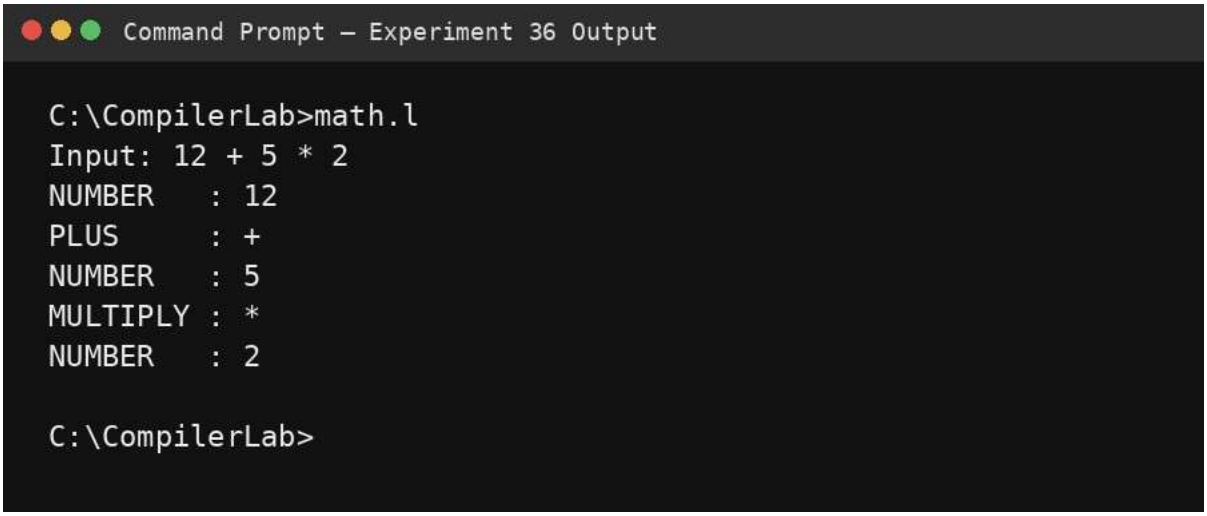
PROGRAM

```
%{
#include <stdio.h>
%}
%%
[0-9]+      { printf("NUMBER   : %s\n",yytext); }
"+"         { printf("PLUS     : +\n"); }
"-"         { printf("MINUS    : -\n"); }
```

AIM

```
"*"        { printf("MULTIPLY : *\n"); }
"/"        { printf("DIVIDE   : /\n"); }
[ \t\n]+   ;
.          { printf("OTHER    : %s\n",yytext); }
%%
int yywrap(void){ return 1; } int
main(void){ yylex(); return 0; }
```

OUTPUT



RESULT

Thus, the basic mathematical operands and operators were recognized successfully.

EXPERIMENT 37: LEX – frequency of a word in a sentence

To count the frequency of a specified word in a sentence using LEX.

ALGORITHM

- Read the word to be counted conceptually as the target token.
- Match alphabetic words from the sentence.
- Compare each matched word with the target.
- Increment the frequency for every exact match.
- Display the final count.

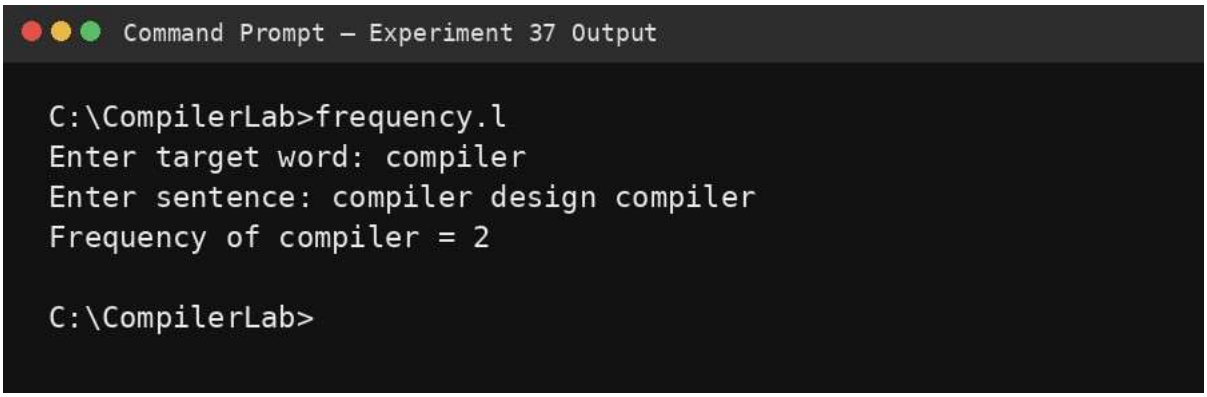
PROGRAM

```
%{
#include <stdio.h>
#include <string.h>
char target[50]; int
count=0;
```

AIM

```
%}
%%
[A-Za-z]+ { if(!strcmp(yytext,target)) count++; }
[ \t\n]+ ;
. ;
%%
int yywrap(void){ return 1; } int
main(void){
   printf("Enter target word: ");
scanf("%49s",target);     getchar();
   printf("Enter sentence: ");
   yylex();
   printf("Frequency of %s = %d\n",target,count);
return 0;
}
```

OUTPUT



RESULT

Thus, the frequency of the specified word was counted successfully.

EXPERIMENT 38: LEX – length of the longest word

To find the length of the longest word in an input text using LEX.

ALGORITHM

- Match each alphabetic word.
- Compare its length with the current maximum.
- Store the word when a new maximum is found.
- After scanning, print the longest word and its length.

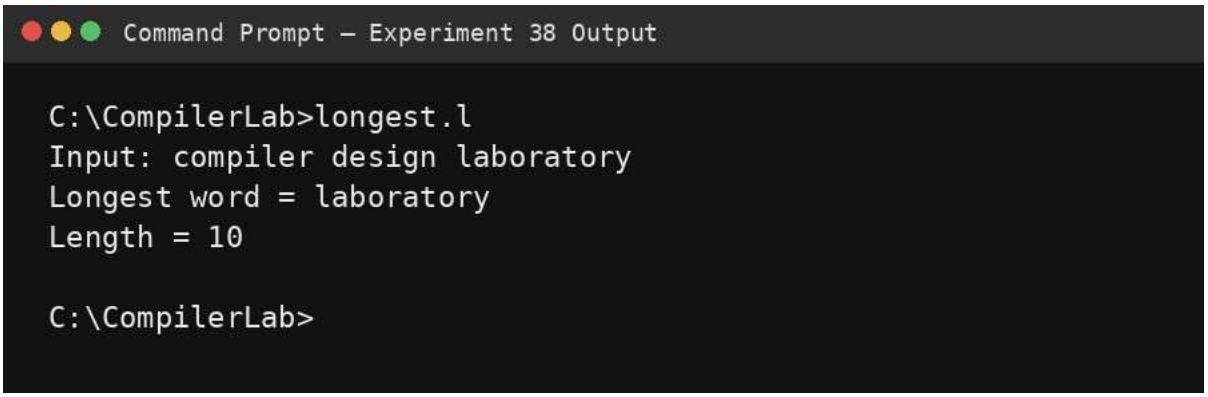
PROGRAM

```
%{
#include <stdio.h> #include
<string.h>
char longest[100]="";
%}
```

AIM

```
%%
[A-Za-z]+ {
   if(yyleng > (int)strlen(longest)) strcpy(longest,yytext); }
[ \t\n]+ ;
. ;
%%
int yywrap(void){ return 1; }
int main(void){     yylex();
   printf("Longest word = %s\nLength = %zu\n",longest,strlen(longest));
return 0;
}
```

OUTPUT



RESULT

Thus, the longest word and its length were obtained successfully.

EXPERIMENT 39: LEX – replace one word with another

To replace every occurrence of one word with another word in a file.

ALGORITHM

- Specify the source word and replacement word.
- Match alphabetic words from the input.
- If the word equals the source word, print the replacement.
- Otherwise print the original word.
- Write or display the resulting text.

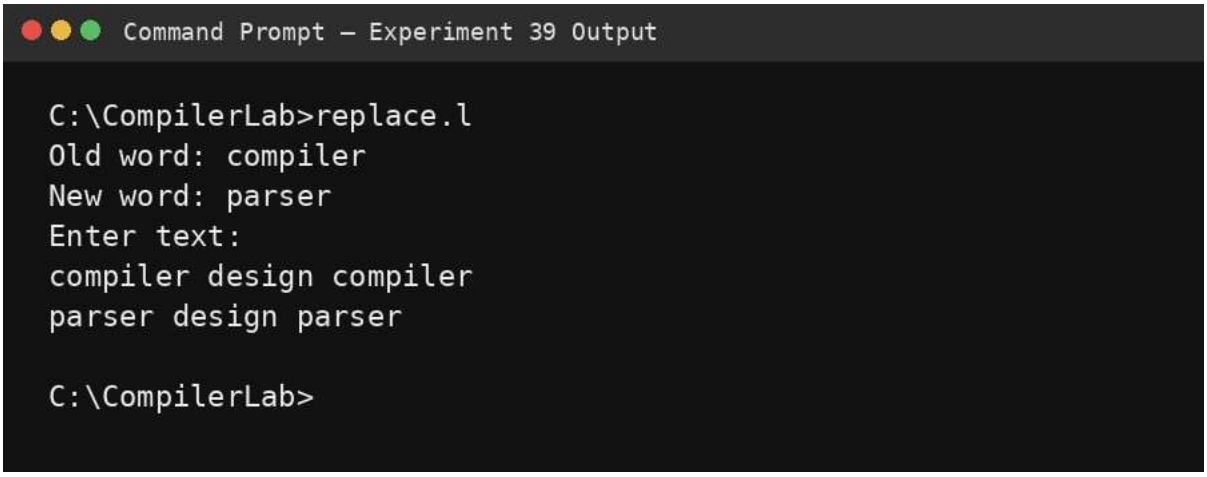
PROGRAM

```
%{
#include <stdio.h> #include
<string.h>
char oldword[50], newword[50];
%}
```

AIM

```
%%
[A-Za-z]+ {
   if(!strcmp(yytext,oldword)) printf("%s",newword);
else printf("%s",yytext);
}
\n         { printf("\n"); }
.          { ECHO; }
%%
int yywrap(void){ return 1; } int main(void){
printf("Old word: "); scanf("%49s",oldword);
printf("New word: "); scanf("%49s",newword);
getchar();
   printf("Enter text:\n");
yylex();
   return 0;
}
```

OUTPUT



RESULT

Thus, every occurrence of the selected word was replaced by the specified replacement word.

EXPERIMENT 40: FLEX – token separation for a C program

To implement a FLEX scanner that separates tokens in a C program and displays each token with an appropriate caption.

ALGORITHM

- Define token patterns for C keywords, identifiers, numeric constants, strings, operators and delimiters.
- Scan the input C program.
- Print each token and its caption.

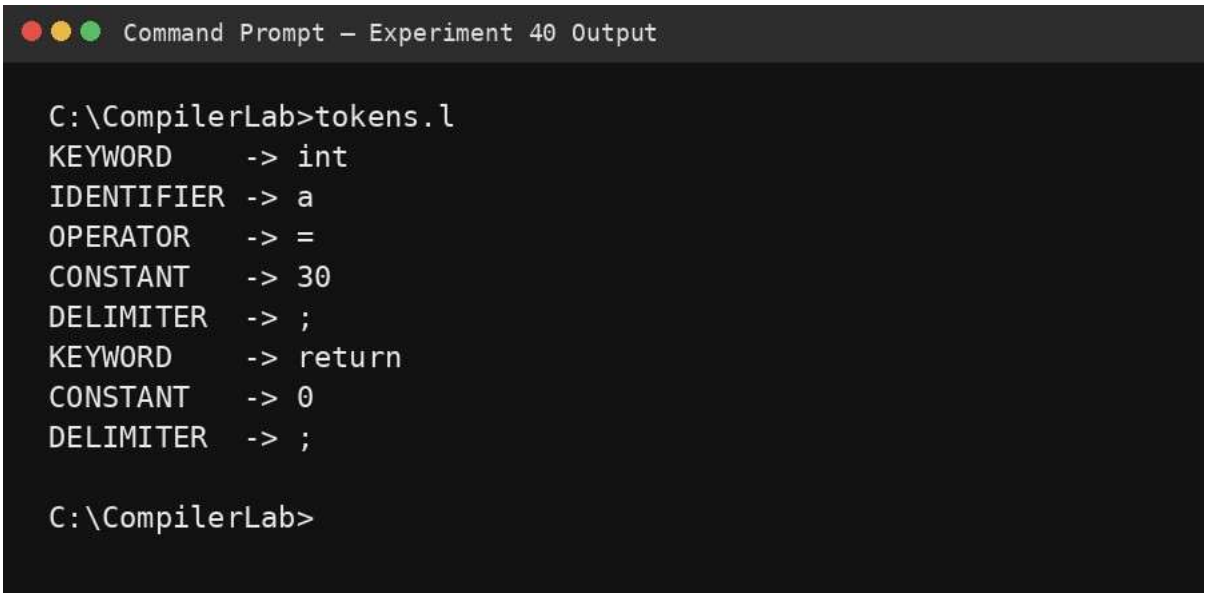
PROGRAM

```
%{
#include <stdio.h>
%}
%%
"int"|"char"|"float"|"double"|"void"|"return"|"if"|"else"|"for"|"while"
```

AIM

```
        { printf("KEYWORD    -> %s\n",yytext); }
[A-Za-z_][A-Za-z0-9_]* { printf("IDENTIFIER -> %s\n",yytext); }
[0-9]+(\.[0-9]+)?      { printf("CONSTANT   -> %s\n",yytext); }
\"([^\\\"]|\\.)*\"     { printf("STRING     -> %s\n",yytext); }
"=="|"!="|"<="|">="|"+"|"-"|"*"|"/"|"="
        { printf("OPERATOR   -> %s\n",yytext); }
[(){};,]              { printf("DELIMITER  -> %s\n",yytext); }
"//"[^\n]*            ;
"/*"([^*]|\*+[^*/])*\*+"/" ;
[ \t\n]+              ;
.                     { printf("OTHER      -> %s\n",yytext); }
%%
int yywrap(void){ return 1; } int
main(void){ yylex(); return 0; }
```

OUTPUT



RESULT

Thus, the FLEX scanner successfully separated the tokens of a C program and displayed appropriate captions.